SPHERES AND HOMOTOPY THEORY

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Abstract. The homotopy groups of spheres are among the simplest objects to define in algebraic topology and among the hardest to compute. This exposition uses them as a guide through several basic techniques of homotopy theory: covering spaces, cellular approximation, fibrations and their long exact sequences, and suspension. We show how each technique computes part of the answer. Along the way we prove that $\pi_n(S^1) = 0$ for $n \geq 2$, that S^n is $(n-1)$-connected, that $\pi_n(S^n) \cong \mathbb{Z}$, and that $\pi_3(S^2) \cong \mathbb{Z}$ is generated by the Hopf map. We also prove that $\pi_i(S^4) \cong \pi_i(S^7) \oplus \pi_{i-1}(S^3)$ for $i \geq 2$, and that $\pi_{n+1}(S^n)$ is cyclic for $n \geq 3$. We prove the Freudenthal suspension theorem from the homotopy excision theorem, and use it to introduce stable homotopy groups. We prove that they form a generalized homology theory, establish a stable Hurewicz theorem, and describe the ring of stable stems. We close with an introduction to stable homotopy theory: the Spanier–Whitehead category and spectra, the Pontryagin–Thom description of the stems by framed manifolds, the Hopf invariant and the J-homomorphism, the Adams spectral sequence, and the chromatic picture.

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1. Introduction

Much of mathematics is organized around classification problems: given a class of objects, describe them all up to an appropriate notion of equivalence. Sets are classified up to bijection by their cardinality, and finitely generated abelian groups up to isomorphism by their rank and torsion. In topology the natural problem is to classify spaces up to homeomorphism. In this generality the problem is hopeless. Markov proved that there is no algorithm that decides whether two closed 4-manifolds are homeomorphic [Mar58].

One therefore asks a more modest question: given two spaces X and Y, how can we *prove* that they are not homeomorphic? The standard approach is to find *invariants*, properties or quantities preserved by homeomorphisms. Compactness, connectedness, and the Hausdorff property are basic examples. Algebraic topology attaches to each space an algebraic object, such as a group, in a way that respects continuous maps. Such invariants can distinguish $\mathbb{R}^2$ from $\mathbb{R}^3$, a sphere from a torus, and S^2 from S^3, none of which the point-set invariants can do.

The homotopy groups $\pi_n(X)$ are the most fundamental of these invariants. They are easy to define: $\pi_n(X)$ is the set of homotopy classes of maps from the n-sphere into X. Yet even for the simplest possible target, a sphere, they are exceedingly hard to compute. The groups $\pi_i(S^n)$ are known only in a finite range, and the attempt to compute them has driven the development of much of modern algebraic topology. Our goal is to use these groups as a guide to several foundational techniques, and to show both what each technique can compute and where it stops.

Outline.

- Section 2 recalls homotopy groups and relative homotopy groups and the long exact sequence of a pair.
- Section 3 uses covering spaces to show that a covering map induces isomorphisms on π_n for $n \geq 2$. In particular all homotopy groups of S^1 are computed.
- Section 4 uses cellular approximation to show that S^n is $(n-1)$-connected.
- Section 5 introduces fibrations and proves their long exact sequence. Applied to the Hopf fibrations, it gives $\pi_2(S^2) \cong \mathbb{Z}$, $\pi_i(S^2) \cong \pi_i(S^3)$ for $i \geq 3$, and $\pi_i(S^4) \cong \pi_i(S^7) \oplus \pi_{i-1}(S^3)$.
- Section 6 studies suspension and proves the Freudenthal suspension theorem, which gives $\pi_n(S^n) \cong \mathbb{Z}$, $\pi_3(S^2) \cong \mathbb{Z}$, and the stability of the groups $\pi_{n+k}(S^n)$ for n large.
- Section 7 defines stable homotopy groups and proves that they form a generalized homology theory. It then describes the stable stems and the ring they form.

- Section 8 is an introduction to stable homotopy theory. It covers spectra, framed manifolds, the Hopf invariant and the J-homomorphism, the Adams spectral sequence, and the chromatic picture. It explains many of the patterns visible in the tables of Section 7.

Prerequisites and conventions. We assume a first graduate course in algebra [Hun74, Lan02] and a first course in algebraic topology, including the fundamental group, covering spaces, and CW complexes [Hat02, May99]. Unless otherwise stated, spaces are pointed and maps and homotopies preserve basepoints; we write $\mathbf{Top}_*$ for the category of pointed spaces. A CW complex is always given a 0-cell as its basepoint. We write $I = [0, 1]$.

2. Homotopy Groups

2.1. **Absolute homotopy groups.** Let I^n be the n-dimensional cube and ∂I^n its boundary, the set of points with at least one coordinate equal to 0 or 1. For a pointed space (X, x_0) and $n \geq 0$, define $\pi_n(X, x_0)$ to be the set of homotopy classes of maps

$$f\colon (I^n, \partial I^n) \to (X, x_0),$$

where the homotopies are required to send ∂I^n to x_0 at all times. (For $n = 0$ we take I^0 to be a point and $\partial I^0 = \varnothing$, so $\pi_0(X, x_0)$ is the set of path components of X, pointed by the component of x_0.) The quotient $I^n/\partial I^n$ is homeomorphic to S^n, so equivalently

$$\pi_n(X, x_0) = [(S^n, *), (X, x_0)]$$

is the set of based homotopy classes of based maps $S^n \to X$. For $n = 1$ this is the fundamental group.

For $n \geq 1$, $\pi_n(X, x_0)$ is a group under the operation of concatenation in the first coordinate,

$$(f + g)(s_1, s_2, \ldots, s_n) = \begin{cases} f(2s_1, s_2, \ldots, s_n), & s_1 \in [0, \tfrac{1}{2}], \\ g(2s_1 - 1, s_2, \ldots, s_n), & s_1 \in [\tfrac{1}{2}, 1]. \end{cases}$$

The identity element is the constant map, and the inverse of $[f]$ is represented by $f(1 - s_1, s_2, \ldots, s_n)$; see Figure 1.

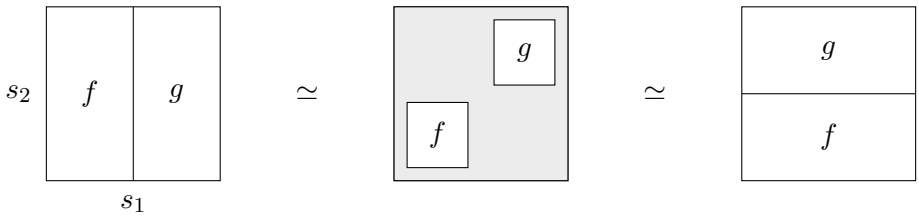


Figure 1. The sum $f + g$ in $\pi_2(X)$ is homotopic to the sum taken in the second coordinate: shrink f and g to small boxes (the shaded region maps to the basepoint), move them apart vertically, and slide them horizontally. Moving them on to swap left and right shows $f + g \simeq g + f$.

Proposition 2.1. *For $n \geq 2$ the group $\pi_n(X, x_0)$ is abelian.*

Proof. Since f and g are constant on the boundary of their cubes, we may shrink the domains of f and g in $f + g$ to smaller cubes, mapping the complement to x_0. When $n \geq 2$ there is a second coordinate direction, so the two small cubes can be slid around each other until

their positions are exchanged, as in Figure 1. Expanding the cubes again gives a homotopy $f + g \simeq g + f$. □

Remark 2.2. For $n = 1$ there is no room to move the two cubes past each other, and indeed π_1 can be nonabelian.

We record the basic formal properties; see [Hat02, §4.1].

(1) *Functoriality.* A based map $\varphi\colon X \to Y$ induces a homomorphism $\varphi_*\colon \pi_n(X, x_0) \to \pi_n(Y, \varphi(x_0))$, $[f] \mapsto [\varphi \circ f]$, and $(\psi\varphi)_* = \psi_*\varphi_*$.
(2) *Homotopy invariance.* Based homotopic maps induce the same homomorphism. A homotopy equivalence induces isomorphisms on all π_n; in particular, all homotopy groups of a contractible space vanish.
(3) *Change of basepoint.* A path γ from x_0 to x_1 induces an isomorphism $\pi_n(X, x_1) \cong \pi_n(X, x_0)$. Thus when X is path-connected the isomorphism class of $\pi_n(X, x_0)$ does not depend on x_0, and we simply write $\pi_n(X)$.
(4) *Products.* The projections induce an isomorphism $\pi_n(X \times Y) \cong \pi_n(X) \times \pi_n(Y)$.

Definition 2.3. A space X is *n-connected* if $\pi_i(X, x_0) = 0$ for $0 \le i \le n$ and every choice of x_0. So 0-connected means path-connected, and 1-connected means simply connected.

2.2. **Relative homotopy groups.** The long exact sequences of Sections 5 and 6 are built from relative homotopy groups, which we now recall. Regard I^{n-1} as the face $\{s_n = 0\}$ of I^n, and let J^{n-1} be the closure of $\partial I^n \setminus I^{n-1}$, that is, the union of all the other faces.

Definition 2.4. Let $A \subseteq X$ with $x_0 \in A$, and let $n \ge 1$. The *relative homotopy set* $\pi_n(X, A, x_0)$ is the set of homotopy classes of maps

$$f\colon (I^n, \partial I^n, J^{n-1}) \to (X, A, x_0),$$

with homotopies through maps of the same form.

Thus f sends the face I^{n-1} into A and all other faces to the basepoint. For $n \ge 2$ the formula above (concatenation in s_1) makes $\pi_n(X, A, x_0)$ a group, abelian for $n \ge 3$; $\pi_1(X, A, x_0)$ is only a pointed set. A map $(I^n, \partial I^n) \to (X, x_0)$ is in particular a map of triples, and restricting a map of triples to the face I^{n-1} gives a map $(I^{n-1}, \partial I^{n-1}) \to (A, x_0)$. These define maps $j_*\colon \pi_n(X, x_0) \to \pi_n(X, A, x_0)$ and $\partial\colon \pi_n(X, A, x_0) \to \pi_{n-1}(A, x_0)$.

Theorem 2.5 ([Hat02, Thm. 4.3])**.** *For a pointed pair (X, A) the sequence*

$$\cdots \to \pi_n(A) \xrightarrow{i_*} \pi_n(X) \xrightarrow{j_*} \pi_n(X, A) \xrightarrow{\partial} \pi_{n-1}(A) \to \cdots \to \pi_1(X, A) \xrightarrow{\partial} \pi_0(A) \xrightarrow{i_*} \pi_0(X)$$

is exact. (Near the right end, where the terms are only pointed sets, exactness means that the image of each map equals the preimage of the basepoint under the next.)

Definition 2.6. A pair (X, A) is called *n-connected* if $\pi_0(A) \to \pi_0(X)$ is surjective and $\pi_i(X, A, x_0) = 0$ for $1 \le i \le n$ and all $x_0 \in A$.

Example 2.7. If X is contractible, then $\pi_n(X) = 0$ for all n, so by exactness $\partial\colon \pi_n(X, A) \to \pi_{n-1}(A)$ is a bijection for every $n \ge 1$. In particular, if A is $(n-1)$-connected and X is contractible, the pair (X, A) is n-connected.

3. Covering Spaces

Covering spaces are the main tool for computing fundamental groups. We now see what they say about the higher homotopy groups. The answer is that a covering map is invisible to π_n for $n \geq 2$. This settles $\pi_n(S^1)$ completely, but it also shows that covering spaces cannot help with the higher homotopy groups of S^n for $n \geq 2$.

Theorem 3.1. *Let $p\colon (E, e_0) \to (B, b_0)$ be a covering map. Then $p_*\colon \pi_n(E, e_0) \to \pi_n(B, b_0)$ is an isomorphism for every $n \geq 2$, and injective for $n = 1$.*

Proof. Surjectivity for $n \geq 2$. Let $f\colon (S^n, *) \to (B, b_0)$ be a based map. Since $n \geq 2$, the sphere S^n is simply connected and locally path-connected. By the lifting criterion for covering spaces [Hat02, Prop. 1.33], f lifts to a map $\tilde{f}\colon (S^n, *) \to (E, e_0)$ with $p\tilde{f} = f$. Hence $[f] = p_*[\tilde{f}]$.

Injectivity for $n \geq 1$. Let $\tilde{f}\colon (S^n, *) \to (E, e_0)$ be such that $p\tilde{f}$ is null-homotopic, via a based homotopy $H\colon S^n \times I \to B$ from $p\tilde{f}$ to the constant map. By the homotopy lifting property of covering spaces, H lifts to a homotopy $\tilde{H}\colon S^n \times I \to E$ starting at $\tilde{H}_0 = \tilde{f}$. Two observations show that $\tilde{H}$ is a based null-homotopy of $\tilde{f}$:

- the path $t \mapsto \tilde{H}(*, t)$ lifts the constant path at b_0 and starts at e_0, so by uniqueness of path lifting it is constant at e_0;
- the map $\tilde{H}_1$ lifts the constant map at b_0, so it takes values in the discrete fibre $p^{-1}(b_0)$. Since S^n is connected, $\tilde{H}_1$ is constant, and its value is $\tilde{H}_1(*) = e_0$.

Hence $[\tilde{f}] = 0$, and p_* is injective because it is a homomorphism. □

Corollary 3.2. *If the universal cover of a path-connected space X is contractible, then $\pi_n(X) = 0$ for all $n \geq 2$.*

Example 3.3. The exponential map $\mathbb{R} \to S^1$ is a universal cover with contractible total space, and $\pi_1(S^1) \cong \mathbb{Z}$. Therefore

$$\pi_n(S^1) \cong \begin{cases} \mathbb{Z}, & n = 1, \\ 0, & n \geq 2. \end{cases}$$

This is the first complete calculation of all the homotopy groups of a sphere. In the same way, the torus $T^k = \mathbb{R}^k/\mathbb{Z}^k$ has $\pi_1(T^k) \cong \mathbb{Z}^k$ and $\pi_n(T^k) = 0$ for $n \geq 2$.

Example 3.4. For $n \geq 2$ the quotient map $S^n \to \mathbb{R}P^n$ is a two-sheeted covering, so $\pi_k(\mathbb{R}P^n) \cong \pi_k(S^n)$ for all $k \geq 2$. Once we know $\pi_n(S^n) \cong \mathbb{Z}$ (Corollary 6.7), this gives $\pi_n(\mathbb{R}P^n) \cong \mathbb{Z}$, even though $H_n(\mathbb{R}P^n; \mathbb{Z}) = 0$ when n is even.

Theorem 3.1 also shows the limitation of the method. Since S^n is simply connected for $n \geq 2$, its only connected covering is the identity, and covering spaces say nothing about its homotopy groups. What we need is a generalization of covering spaces in which the fibre is allowed to be a space with interesting homotopy, rather than a discrete set. This is the idea of a fibration, which we take up in Section 5. First, we need a tool that computes the homotopy groups of S^n in degrees below n.

4. Cellular Approximation

Computing $\pi_i(Y)$ means understanding all continuous maps $S^i \to Y$, and continuous maps can be quite wild: there are surjective maps $S^1 \to S^2$ (space-filling curves). Cellular approximation says that, up to homotopy, maps between CW complexes can be made to

respect the cell structures. This is enough to compute the homotopy groups of S^n below degree n.

Definition 4.1. A map $f\colon X \to Y$ of CW complexes is *cellular* if $f(X^{(k)}) \subseteq Y^{(k)}$ for every k, where $X^{(k)}$ denotes the k-skeleton.

Theorem 4.2 (Cellular approximation [Hat02, Thm. 4.8])**.** *Every map $f\colon X \to Y$ of CW complexes is homotopic to a cellular map. If f is already cellular on a subcomplex $A \subseteq X$, the homotopy may be taken to be stationary on A.*

Remark 4.3. Some authors, following Steenrod and Miller [Mil21], call such maps *skeletal*. The key input to the proof is the following lemma. If $k < n$, a map $D^k \to X \cup e^n$ can be pushed off a point of the open n-cell by a homotopy that is fixed on the part of D^k already mapping into X. One then deforms the image radially into X and inducts over the cells of the domain.

The n-sphere has a CW structure with exactly two cells, $S^n = e^0 \cup e^n$. The n-cell is attached by the constant map $\partial D^n = S^{n-1} \to e^0$, so $S^n = D^n/\partial D^n$. In particular, the k-skeleton of S^n is the single point e^0 for every $k < n$.

Corollary 4.4. *The sphere S^n is $(n-1)$-connected: $\pi_i(S^n) = 0$ for all $i < n$.*

Proof. For $n \geq 1$, S^n is path-connected, so $\pi_0(S^n)$ is trivial. Let $1 \leq i < n$ and let $f\colon S^i \to S^n$ be a based map. Give both spheres their two-cell structures. The map f sends the 0-cell to the 0-cell, so it is cellular on the subcomplex e^0. By Theorem 4.2, f is homotopic relative to e^0 (that is, by a based homotopy) to a cellular map g. But

$$g(S^i) = g\big((S^i)^{(i)}\big) \subseteq (S^n)^{(i)} = e^0,$$

so g is constant. Hence $[f] = 0$ in $\pi_i(S^n)$. □

The same argument proves a more general statement. If X is a CW complex, then the inclusion of the k-skeleton induces an isomorphism $\pi_i(X^{(k)}) \to \pi_i(X)$ for $i < k$ and a surjection for $i = k$ [Hat02, Cor. 4.12]. So $\pi_i(X)$ is determined by the $(i+1)$-skeleton of X.

Cellular approximation tells us where the homotopy groups of S^n *start*, but nothing about $\pi_n(S^n)$ or beyond. The next section supplies the first tool that sees beyond this range.

5. Fibrations

A covering map $p\colon E \to B$ is characterized, up to point-set details, by unique lifting of homotopies. A fibration keeps the existence of lifts but drops uniqueness, so its fibres can be arbitrary spaces rather than discrete sets. The payoff is a long exact sequence relating the homotopy groups of the fibre, the total space, and the base. It is one of the most important computational tools in the subject.

5.1. Definitions.

Definition 5.1. A map $p\colon E \to B$ has the *homotopy lifting property* (HLP) with respect to a space X if for every map $\tilde{h}_0\colon X \to E$ and every homotopy $h\colon X \times I \to B$ with $h(x,0) = p\tilde{h}_0(x)$, there is a homotopy $\tilde{h}\colon X \times I \to E$ with $\tilde{h}(x,0) = \tilde{h}_0(x)$ and $p\tilde{h} = h$:

$$\begin{array}{ccc} X \times \{0\} & \xrightarrow{\tilde{h}_0} & E \\ \downarrow & \overset{\tilde{h}}{\nearrow} & \downarrow p \\ X \times I & \xrightarrow[h]{} & B. \end{array}$$

The map p is a *Hurewicz fibration* if it has the HLP with respect to all spaces, and a *Serre fibration* if it has the HLP with respect to all cubes I^k, $k \geq 0$.

Every Hurewicz fibration is a Serre fibration. Serre fibrations are all we need for homotopy groups, and they include the most important examples.

Definition 5.2. A *fibre bundle* with fibre F is a map $p\colon E \to B$ such that every point of B has an open neighbourhood U and a homeomorphism $\varphi\colon p^{-1}(U) \to U \times F$ with $\mathrm{pr}_1 \circ \varphi = p$.

Proposition 5.3 ([Hat02, Prop. 4.48])**.** *Every fibre bundle is a Serre fibration.*

A covering map is exactly a fibre bundle with discrete fibre. We will need the following relative form of the lifting property.

Lemma 5.4. *Let $p\colon E \to B$ be a Serre fibration and let $J^k \subseteq I^{k+1}$ be the union of all faces of I^{k+1} but one. Then every map $h\colon I^{k+1} \to B$, together with a lift of $h|_{J^k}$ to E, admits a lift $\tilde{h}\colon I^{k+1} \to E$ extending the given one.*

Proof. There is a homeomorphism of pairs $(I^{k+1}, J^k) \cong (I^k \times I,\ I^k \times \{0\})$; for instance, project radially from a point just beyond the centre of the omitted face. Under this homeomorphism, the statement becomes the HLP for I^k. □

5.2. **The long exact sequence.** Fix a Serre fibration $p\colon E \to B$, a basepoint $b_0 \in B$, the fibre $F = p^{-1}(b_0)$, and a basepoint $e_0 \in F$. Since $p(F) = b_0$, the map p induces maps

$$p_*\colon \pi_n(E, F, e_0) \to \pi_n(B, b_0, b_0) = \pi_n(B, b_0).$$

Theorem 5.5. *For every $n \geq 1$, $p_*\colon \pi_n(E, F, e_0) \to \pi_n(B, b_0)$ is a bijection. Consequently there is a long exact sequence*

$$\cdots \to \pi_n(F) \to \pi_n(E) \xrightarrow{p_*} \pi_n(B) \xrightarrow{\partial} \pi_{n-1}(F) \to \cdots \to \pi_1(B) \xrightarrow{\partial} \pi_0(F) \to \pi_0(E) \to \pi_0(B),$$

where ∂ is the composite of $(p_*)^{-1}$ with the boundary map of the pair (E, F).

Proof. Surjectivity. Let $f\colon (I^n, \partial I^n) \to (B, b_0)$. Lift $f|_{J^{n-1}}$ to the constant map at e_0. By Lemma 5.4, this extends to a lift $g\colon I^n \to E$ of f. Since $f(\partial I^n) = b_0$, the face I^{n-1} is sent by g into $p^{-1}(b_0) = F$. So g represents a class in $\pi_n(E, F, e_0)$ with $p_*[g] = [f]$.

Injectivity. Let $g_0, g_1\colon (I^n, \partial I^n, J^{n-1}) \to (E, F, e_0)$, and let $G\colon I^n \times I \to B$ be a homotopy from pg_0 to pg_1 through maps $(I^n, \partial I^n) \to (B, b_0)$. Define a lift of G on the subspace

$$I^n \times \{0\} \ \cup\ I^n \times \{1\} \ \cup\ J^{n-1} \times I$$

by g_0, g_1, and the constant map e_0. This subspace is the union of all faces of I^{n+1} except $I^{n-1} \times I$, so Lemma 5.4 extends the lift to $\tilde{G}\colon I^n \times I \to E$. Since $G(\partial I^n \times I) = b_0$, the remaining face $I^{n-1} \times I$ is sent into F, and $\tilde{G}$ is a homotopy from g_0 to g_1 through maps of triples. Hence $[g_0] = [g_1]$.

The long exact sequence. Substitute $\pi_n(B)$ for $\pi_n(E, F)$ in the long exact sequence of the pair (E, F) (Theorem 2.5). The composite $\pi_n(E) \to \pi_n(E, F) \cong \pi_n(B)$ is p_*. This gives exactness everywhere except at $\pi_0(E)$. There, a path component of E maps to the component of b_0 if and only if it contains a point e joined to b_0 by a path γ in B. Lifting γ starting at e produces a path in E from e to a point of F, so the component meets F. □

Remark 5.6. If B is path-connected, $\pi_0(B)$ is a single point and the sequence ends with $\pi_1(B) \to \pi_0(F) \to \pi_0(E) \to 0$, the last map being surjective. If in addition E is path-connected, $\partial\colon \pi_1(B) \to \pi_0(F)$ is surjective.

Example 5.7. Applied to a covering map, where F is discrete so that $\pi_n(F) = 0$ for $n \geq 1$, Theorem 5.5 recovers Theorem 3.1. It also gives an exact sequence $0 \to \pi_1(E) \to \pi_1(B) \to \pi_0(F)$, which identifies the fibre with the cosets of $p_*\pi_1(E)$ when E is connected. So fibrations genuinely generalize covering spaces.

The following observation turns the long exact sequence into a direct sum decomposition, and will be used for the quaternionic Hopf fibration below. Recall that the reduced suspension of a pointed space F is $\Sigma F = F \times I/(F \times \partial I \cup \{x_0\} \times I)$; we study it in detail in Section 6.

Proposition 5.8. *In the situation of Theorem 5.5, suppose E and F are path-connected and the inclusion $i\colon F \hookrightarrow E$ is based null-homotopic. Then for every $n \geq 2$,*

$$\pi_n(B) \cong \pi_n(E) \oplus \pi_{n-1}(F).$$

Proof. Since $i_* = 0$ in every degree, the long exact sequence breaks into short exact sequences

$$0 \to \pi_n(E) \xrightarrow{p_*} \pi_n(B) \xrightarrow{\partial} \pi_{n-1}(F) \to 0.$$

For $n \geq 2$, $\pi_n(B)$ is abelian, so it suffices to construct a homomorphism $\sigma\colon \pi_{n-1}(F) \to \pi_n(B)$ with $\partial\sigma = \mathrm{id}$.

Let $H\colon F \times I \to E$ be a based null-homotopy with $H_0 = i$ and H_1 constant at e_0. Then pH sends $F \times \{0\}$, $F \times \{1\}$, and $\{e_0\} \times I$ to b_0, so it descends to a based map $\varphi\colon \Sigma F \to B$. Given $f\colon (I^{n-1}, \partial I^{n-1}) \to (F, e_0)$, let $\Sigma f\colon I^n \to \Sigma F$ be $(s,t) \mapsto [f(s), t]$, and set $\sigma[f] = [\varphi \circ \Sigma f]$. The map σ is a homomorphism, since both Σ and φ_* are.

To compute $\partial\sigma[f]$, note that $g(s,t) = H(f(s),t)$ is a lift of $\varphi \circ \Sigma f$ to E. On the face $t = 0$, $g(s,0) = f(s) \in F$. On the face $t = 1$, g is constant at e_0. On $\partial I^{n-1} \times I$, $g = H(e_0, t) = e_0$. So g represents a class in $\pi_n(E, F)$ with $p_*[g] = \sigma[f]$ and boundary $[g|_{I^{n-1}}] = [f]$. Thus $\partial\sigma[f] = [f]$. □

5.3. **The Hopf fibrations.** We now apply the long exact sequence to the spheres. The key examples come from the normed division algebras $\mathbb{C}$ and $\mathbb{H}$.

Example 5.9 (The complex Hopf fibration)**.** Regard S^3 as the unit sphere in $\mathbb{C}^2$ and S^2 as the complex projective line $\mathbb{C}P^1 = \mathbb{C} \cup \{\infty\}$. The *Hopf map* is

$$\eta\colon S^3 \to S^2, \qquad \eta(z_0, z_1) = [z_0 : z_1].$$

The fibre over $[z_0 : z_1]$ is $\{(\lambda z_0, \lambda z_1) : |\lambda| = 1\} \cong S^1$. The map is locally trivial. Over $U_0 = \{z_0 \neq 0\}$, with coordinate $w = z_1/z_0$, the maps

$$\eta^{-1}(U_0) \to U_0 \times S^1, \quad (z_0, z_1) \mapsto \left(w, \tfrac{z_0}{|z_0|}\right), \qquad (w, \lambda) \mapsto \frac{\lambda\,(1, w)}{\sqrt{1 + |w|^2}}$$

are mutually inverse homeomorphisms. The same formulas with the coordinates exchanged work over $U_1 = \{z_1 \neq 0\}$. So η is a fibre bundle with fibre S^1, hence a Serre fibration:

$$S^1 \to S^3 \xrightarrow{\eta} S^2.$$

Theorem 5.10. *$\pi_2(S^2) \cong \mathbb{Z}$, and $\eta_*\colon \pi_i(S^3) \to \pi_i(S^2)$ is an isomorphism for all $i \geq 3$.*

Proof. Consider the long exact sequence of the Hopf fibration:

$$\cdots \to \pi_i(S^1) \to \pi_i(S^3) \xrightarrow{\eta_*} \pi_i(S^2) \xrightarrow{\partial} \pi_{i-1}(S^1) \to \pi_{i-1}(S^3) \to \cdots.$$

By Corollary 4.4, $\pi_1(S^3) = \pi_2(S^3) = 0$, so $\partial\colon \pi_2(S^2) \to \pi_1(S^1) \cong \mathbb{Z}$ is an isomorphism. For $i \geq 3$, both $\pi_i(S^1)$ and $\pi_{i-1}(S^1)$ vanish by Example 3.3, so η_* is an isomorphism. □

This is the first place where we see that the homotopy groups of S^n do not stop at degree n. As soon as we know $\pi_3(S^3) \cong \mathbb{Z}$, which we prove in Section 6, we will conclude that $\pi_3(S^2) \cong \mathbb{Z}$, generated by $[\eta]$. Hopf discovered this in 1931 [Hop31]; it was the first nonzero homotopy group of a sphere found above its dimension. Theorem 5.10 also shows that S^2 and S^3 have the same homotopy groups in every degree ≥ 3, so that, for example, $\pi_4(S^2) \cong \pi_4(S^3)$.

Example 5.11 (The quaternionic Hopf fibration)**.** Replace $\mathbb{C}$ by the quaternions $\mathbb{H}$. Let $S^7 \subseteq \mathbb{H}^2$ be the unit sphere and let $\mathbb{H}P^1$ be the space of quaternionic lines $\{(q_0\lambda, q_1\lambda) : \lambda \in \mathbb{H}\}$. Then $\mathbb{H}P^1 = \mathbb{H} \cup \{\infty\} \cong S^4$ via $[q_0 : q_1] \mapsto q_1 q_0^{-1}$; this coordinate is unchanged when (q_0, q_1) is replaced by $(q_0\lambda, q_1\lambda)$. The map $S^7 \to S^4$, $(q_0, q_1) \mapsto [q_0 : q_1]$, has fibres $\{(q_0\lambda, q_1\lambda) : |\lambda| = 1\} \cong S^3$. The same local trivializations as in Example 5.9, namely $(q_0, q_1) \mapsto (q_1 q_0^{-1}, q_0/|q_0|)$ with inverse $(w, \lambda) \mapsto (1, w)\lambda/\sqrt{1 + |w|^2}$, show that it is a fibre bundle:

$$S^3 \to S^7 \to S^4.$$

Theorem 5.12. *For all $i \geq 2$,*

$$\pi_i(S^4) \cong \pi_i(S^7) \oplus \pi_{i-1}(S^3).$$

Proof. By Corollary 4.4, $\pi_3(S^7) = 0$, so the inclusion of the fibre $S^3 \hookrightarrow S^7$ is based null-homotopic. Now apply Proposition 5.8. □

For instance, $\pi_7(S^4) \cong \pi_7(S^7) \oplus \pi_6(S^3)$ contains a copy of $\mathbb{Z}$, so S^4 has an infinite homotopy group above its dimension. Applying Proposition 5.8 to the complex Hopf fibration in the same way recovers Theorem 5.10. There is also an octonionic Hopf fibration $S^7 \to S^{15} \to S^8$ [Hat02, §4.2], which gives $\pi_i(S^8) \cong \pi_i(S^{15}) \oplus \pi_{i-1}(S^7)$.

Remark 5.13. By a celebrated theorem of Adams [Ada60], fibre bundles $S^{n-1} \to S^{2n-1} \to S^n$ exist only for $n = 1, 2, 4, 8$, corresponding to the normed division algebras $\mathbb{R}$, $\mathbb{C}$, $\mathbb{H}$, and the octonions $\mathbb{O}$. So this source of computations runs out quickly.

6. Suspension

The long exact sequence relates homotopy groups of spaces of different dimensions, but it does not by itself compute $\pi_3(S^3)$. For that we need a way to compare $\pi_i(X)$ with π_{i+1} of a space one dimension higher. Suspension provides exactly this.

6.1. Suspension and the suspension homomorphism.

Definition 6.1. The *(unreduced) suspension* of a space X is

$$SX = X \times I \big/ \big(X \times \{0\},\ X \times \{1\}\big),$$

where each of the two ends is collapsed to its own point, the two *cone points*. If (X, x_0) is pointed, its *reduced suspension* is

$$\Sigma X = X \times I \big/ \big(X \times \{0\} \cup X \times \{1\} \cup \{x_0\} \times I\big),$$

pointed by the image of the collapsed set. Equivalently, $\Sigma X \cong X \wedge S^1$.

When X is a CW complex with basepoint a 0-cell, $\{x_0\} \times I$ is a contractible subcomplex of SX. Collapsing it is therefore a homotopy equivalence $SX \xrightarrow{\simeq} \Sigma X$ [Hat02, Prop. 0.17], so for homotopy groups the two suspensions are interchangeable. We use Σ, because it is a functor on pointed spaces: a based map $f\colon X \to Y$ induces $\Sigma f\colon \Sigma X \to \Sigma Y$, $[x, t] \mapsto [f(x), t]$, and based homotopies induce based homotopies.

Proposition 6.2. *For all $n \geq 0$ there is a homeomorphism $\Sigma S^n \cong S^{n+1}$.*

Proof. For $n = 0$, $S^0 = \{x_0, x_1\}$ and $\Sigma S^0 = \{x_1\} \times I/\partial I \cong S^1$. For $n \geq 1$, write $S^n = I^n/\partial I^n$. Then ΣS^n is the quotient of $I^n \times I = I^{n+1}$ by the image of

$$\partial I^n \times I \ \cup \ I^n \times \partial I = \partial I^{n+1},$$

so $\Sigma S^n \cong I^{n+1}/\partial I^{n+1} \cong S^{n+1}$. (The map $I^{n+1} \to (I^n/\partial I^n) \times I$ is a quotient map because I is locally compact Hausdorff, so the composite is a quotient map.) □

Proposition 6.3. *For $i \geq 1$, the assignment $[f] \mapsto [\Sigma f]$ defines a homomorphism*

$$\Sigma_* \colon \pi_i(X) \to \pi_{i+1}(\Sigma X),$$

called the suspension homomorphism. *In cube coordinates, $\Sigma f \colon I^{i+1} \to \Sigma X$ is $(s,t) \mapsto [f(s), t]$.*

Proof. By Proposition 6.2, Σf is a map $S^{i+1} \cong \Sigma S^i \to \Sigma X$, and a based homotopy f_u induces a based homotopy Σf_u, so Σ_* is well defined. The sum in π_i is concatenation in the coordinate s_1, and Σ only adds the new coordinate t. So $\Sigma(f+g)$ and $\Sigma f + \Sigma g$ are literally the same map. □

6.2. **The Freudenthal suspension theorem.** The suspension homomorphism is far from an isomorphism in general. Freudenthal's theorem [Fre38] says that it *is* an isomorphism in a range that grows with the connectivity of X. We deduce it from the following fundamental theorem, which describes the failure of excision for homotopy groups. Unlike homology, homotopy groups do not satisfy excision, but they do in a range.

Theorem 6.4 (Homotopy excision, Blakers–Massey [Hat02, Thm. 4.23])**.** *Let X be a CW complex that is the union of subcomplexes A and B whose intersection $C = A \cap B$ is nonempty and connected. If (A, C) is m-connected and (B, C) is n-connected, then the map $\pi_i(A, C) \to \pi_i(X, B)$ induced by inclusion is an isomorphism for $i < m+n$ and a surjection for $i = m+n$.*

To apply it, split the reduced suspension into two halves. Let C_-X and C_+X be the images in ΣX of $X \times [0, \frac{1}{2}]$ and $X \times [\frac{1}{2}, 1]$. Each is a copy of the reduced cone $CX = X \wedge I$, hence contractible, and

$$\Sigma X = C_+X \cup C_-X, \qquad C_+X \cap C_-X = X \times \{\tfrac{1}{2}\} \cong X.$$

If X is a CW complex, these are subcomplexes of the product CW structure.

Theorem 6.5 (Freudenthal suspension theorem)**.** *Let X be an $(n-1)$-connected CW complex with $n \geq 1$. Then*

$$\Sigma_* \colon \pi_i(X) \to \pi_{i+1}(\Sigma X)$$

is an isomorphism for $i < 2n-1$ and a surjection for $i = 2n-1$.

Proof. Since $C_\pm X$ is contractible and X is $(n-1)$-connected, both pairs $(C_\pm X, X)$ are n-connected (Example 2.7). Consider the diagram

$$\begin{array}{ccc} \pi_{i+1}(C_+X, X) & \xrightarrow{\text{exc}} & \pi_{i+1}(\Sigma X, C_-X) \\ \partial \downarrow \cong & & \cong \uparrow j_* \\ \pi_i(X) & \xrightarrow{\Sigma_*} & \pi_{i+1}(\Sigma X). \end{array}$$

The left map is an isomorphism because C_+X is contractible, and the right map is an isomorphism because C_-X is contractible (Theorem 2.5). By Theorem 6.4 with $m = n$, the top

map is an isomorphism for $i+1<2n$ and a surjection for $i+1=2n$. So it remains to show that the diagram commutes, i.e., that $j_*\Sigma_* = \mathrm{exc}\circ\partial^{-1}$.

Let $f\colon (I^i,\partial I^i)\to(X,x_0)$. Its preimage under ∂ is the "cone on f", $c_f(s,u)=[f(s),\frac{1}{2}+\frac{u}{2}]\in C_+X$. Indeed, c_f sends the face $u=0$ to f in X and all other faces to the basepoint. On the other hand, $j_*\Sigma_*[f]$ is represented by $\Sigma f(s,t)=[f(s),t]$, viewed as a map of triples into $(\Sigma X, C_-X, *)$. The homotopy

$$h_v(s,t)=\big[f(s),\,(1-v)\,t+v\max(t,\tfrac{1}{2})\big],\qquad v\in I,$$

is a homotopy of maps of triples: the face $t=0$ goes to $[f(s),v/2]\in C_-X$, and all other faces go to the basepoint. At $v=1$, the lower half $t\le\frac{1}{2}$ is sent to $X\subseteq C_-X$, and the upper half is c_f after a linear reparametrization of t. Shrinking the lower half to the face $t=0$ is a further homotopy of maps of triples, so $j_*[\Sigma f]=\mathrm{exc}[c_f]$, as claimed. □

We will also need to know how connectivity behaves under suspension.

Lemma 6.6. *Let $n\ge 0$. If X is an $(n-1)$-connected CW complex, then ΣX is n-connected. (For $n=0$ the hypothesis is empty: the suspension of any CW complex is path-connected.)*

Proof. For $n=0$ take $Z=X$. For $n\ge 1$, by [Hat02, Cor. 4.16], X is homotopy equivalent, relative to the basepoint, to a CW complex Z whose only cell of dimension less than n is the basepoint. The reduced suspension ΣZ has the basepoint as its unique 0-cell and one $(k+1)$-cell Σe^k for each k-cell $e^k\neq *$ of Z. So all its other cells have dimension at least $n+1$. By cellular approximation, exactly as in Corollary 4.4, ΣZ is n-connected, and $\Sigma X\simeq\Sigma Z$. □

6.3. **First consequences.** We now reap the rewards.

Corollary 6.7. *For every $n\ge 1$, $\pi_n(S^n)\cong\mathbb{Z}$, generated by the class of the identity map.*

Proof. Under the identification $\Sigma S^n\cong S^{n+1}$, the suspension of id_{S^n} is $\mathrm{id}_{S^{n+1}}$. Consider the sequence of suspension homomorphisms

$$\pi_1(S^1)\xrightarrow{\Sigma_*}\pi_2(S^2)\xrightarrow{\Sigma_*}\pi_3(S^3)\xrightarrow{\Sigma_*}\pi_4(S^4)\xrightarrow{\Sigma_*}\cdots.$$

Since S^n is $(n-1)$-connected, Theorem 6.5 applies with $i=n$. For $n=1$ we have $i=1=2n-1$, so the first map is surjective. For $n\ge 2$ we have $n<2n-1$, so all the subsequent maps are isomorphisms.

The first map is a surjection $\mathbb{Z}\cong\pi_1(S^1)\to\pi_2(S^2)\cong\mathbb{Z}$, using Theorem 5.10 for the target. A surjection $\mathbb{Z}\to\mathbb{Z}$ is an isomorphism. Since $[\mathrm{id}_{S^1}]$ generates $\pi_1(S^1)$, its images $[\mathrm{id}_{S^n}]$ generate $\pi_n(S^n)$ for every n. □

Remark 6.8. The isomorphism $\pi_n(S^n)\cong\mathbb{Z}$ is the *degree*: a map $f\colon S^n\to S^n$ represents $\deg(f)$ times the generator, where $\deg f$ is defined by $f_*=\deg(f)\cdot\mathrm{id}$ on $H_n(S^n;\mathbb{Z})\cong\mathbb{Z}$. This also follows from the Hurewicz theorem [Hat02, Thm. 4.32].

Corollary 6.9. *$\pi_3(S^2)\cong\mathbb{Z}$, generated by the Hopf map η.*

Proof. By Theorem 5.10, $\eta_*\colon\pi_3(S^3)\to\pi_3(S^2)$ is an isomorphism, and it sends the generator $[\mathrm{id}_{S^3}]$ to $[\eta\circ\mathrm{id}]=[\eta]$. □

At this point we have every homotopy group of S^2 through degree 3. The Freudenthal theorem also gives information one step further.

Corollary 6.10. *For $n\ge 3$ the group $\pi_{n+1}(S^n)$ is cyclic, generated by the $(n-2)$-fold suspension $\Sigma^{n-2}\eta$ of the Hopf map. All these groups are isomorphic via suspension, and $\pi_4(S^2)\cong\pi_4(S^3)$.*

Proof. Apply Theorem 6.5 to $X = S^2$ (so $n = 2$) with $i = 3 = 2n-1$. The map $\Sigma_*\colon \pi_3(S^2) \cong \mathbb{Z} \to \pi_4(S^3)$ is surjective, so $\pi_4(S^3)$ is cyclic, generated by $\Sigma\eta$. For $n \geq 3$, $\Sigma_*\colon \pi_{n+1}(S^n) \to \pi_{n+2}(S^{n+1})$ is an isomorphism, since $n+1 < 2n-1$. The last statement is Theorem 5.10. □

In fact $\pi_{n+1}(S^n) \cong \mathbb{Z}/2$ for all $n \geq 3$ [Hat02, §4.L]. Showing that $\Sigma\eta \neq 0$ requires a cohomology operation (the Steenrod square Sq^2 on $\Sigma\mathbb{C}P^2$), and showing that $2\Sigma\eta = 0$ requires a further geometric argument. Both are beyond the tools developed here. This is a recurring theme: the long exact sequence and the suspension theorem tell us which groups are *related*, but determining a group outright usually needs further input.

7. Stable Homotopy Groups

7.1. **Stability.** The Freudenthal theorem says that suspension becomes an isomorphism in a range that grows with connectivity. Each suspension raises connectivity by one (Lemma 6.6), so iterating it eventually makes every map an isomorphism. Concretely, let X be an $(n-1)$-connected CW complex, where $n \geq 0$ (for $n = 0$ this is no condition at all), and fix k. Then $\Sigma^m X$ is $(n+m-1)$-connected, so in the sequence

$$\pi_k(X) \xrightarrow{\Sigma_*} \pi_{k+1}(\Sigma X) \xrightarrow{\Sigma_*} \pi_{k+2}(\Sigma^2 X) \xrightarrow{\Sigma_*} \cdots$$

the map out of $\pi_{k+m}(\Sigma^m X)$ is an isomorphism as soon as $m \geq 1$ and $k+m < 2(n+m)-1$, i.e., for $m > k-2n+1$. The sequence is eventually constant.

Definition 7.1. The *k-th stable homotopy group* of a pointed CW complex X is

$$\pi_k^s(X) = \operatorname{colim}_m \pi_{k+m}(\Sigma^m X), \qquad k \in \mathbb{Z},$$

the colimit along the suspension homomorphisms.

By the discussion above, for *every* pointed CW complex X,

$$\pi_k^s(X) \cong \pi_{k+m}(\Sigma^m X) \qquad \text{for all } m \geq k+2. \tag{7.1}$$

In particular $\pi_k^s(X)$ is an abelian group, since π_{k+m} is abelian once $k+m \geq 2$. Also $\pi_k^s(X) = 0$ for $k < 0$, since $\Sigma^m X$ is $(m-1)$-connected. Stable homotopy groups are functorial for based maps and depend only on the based homotopy class of the map.

7.2. **Stable homotopy is a homology theory.** Homology owes its computability to two properties: excision, which gives the long exact sequence of a CW pair

$$\cdots \to \tilde{H}_k(A) \to \tilde{H}_k(X) \to \tilde{H}_k(X/A) \to \tilde{H}_{k-1}(A) \to \cdots,$$

and additivity, $\tilde{H}_k(X \vee Y) \cong \tilde{H}_k(X) \oplus \tilde{H}_k(Y)$. Homotopy groups have neither property:

- For the pair (D^2, S^1), with $D^2/S^1 = S^2$, the would-be exact segment $\pi_3(D^2) \to \pi_3(S^2) \to \pi_2(S^1)$ reads $0 \to \mathbb{Z} \to 0$.
- By a theorem of Hilton, $\pi_3(S^2 \vee S^2) \cong \mathbb{Z}^3$. Besides the two Hopf maps it contains a third generator, the Whitehead product of the two inclusions $S^2 \hookrightarrow S^2 \vee S^2$.

The central fact of this section is that both defects disappear after stabilization. The key input is the following consequence of the homotopy excision theorem.

Proposition 7.2 ([Hat02, Prop. 4.28])**.** *Let (X, A) be an r-connected CW pair with A an s-connected complex, where $r, s \geq 0$. Then the quotient map induces a map $\pi_i(X, A) \to \pi_i(X/A)$ that is an isomorphism for $i \leq r+s$ and a surjection for $i = r+s+1$.*

Proposition 7.3. *Let (X, A) be a CW pair, with basepoint a 0-cell of A. Then the sequence*

$$\pi_k^s(A) \xrightarrow{i_*} \pi_k^s(X) \xrightarrow{q_*} \pi_k^s(X/A)$$

induced by the inclusion i and the quotient map q is exact.

Proof. Fix $m \geq k+2$, so that by (7.1) all three groups may be computed at stage m. Suspension commutes with quotients, $\Sigma^m X/\Sigma^m A \cong \Sigma^m(X/A)$, so we must show that

$$\pi_{k+m}(\Sigma^m A) \to \pi_{k+m}(\Sigma^m X) \to \pi_{k+m}\big(\Sigma^m X/\Sigma^m A\big)$$

is exact. We estimate the two connectivities needed for Proposition 7.2:

- $\Sigma^m A$ is $(m-1)$-connected by Lemma 6.6, so we can take $s = m-1$;
- every cell of $X \setminus A$ of dimension d gives a cell of $\Sigma^m X \setminus \Sigma^m A$ of dimension $d+m \geq m$, so the pair $(\Sigma^m X, \Sigma^m A)$ is $(m-1)$-connected [Hat02, Cor. 4.12], and we can take $r = m-1$.

Proposition 7.2 therefore says that $\pi_{k+m}(\Sigma^m X, \Sigma^m A) \to \pi_{k+m}(\Sigma^m(X/A))$ is an isomorphism, because $k+m \leq 2m-2$. The quotient map $\Sigma^m X \to \Sigma^m(X/A)$ factors as

$$\pi_{k+m}(\Sigma^m X) \xrightarrow{j_*} \pi_{k+m}(\Sigma^m X, \Sigma^m A) \xrightarrow{\cong} \pi_{k+m}(\Sigma^m(X/A)),$$

so its kernel equals $\ker j_*$. That kernel is the image of $\pi_{k+m}(\Sigma^m A)$, by the long exact sequence of the pair (Theorem 2.5). □

Theorem 7.4. *The functors π_k^s, $k \in \mathbb{Z}$, on pointed CW complexes have the following properties.*

(1) Homotopy invariance: *based homotopic maps induce the same homomorphisms.*
(2) Suspension: *there are natural isomorphisms $\pi_k^s(X) \cong \pi_{k+1}^s(\Sigma X)$.*
(3) Exactness: *a CW pair (X, A) induces a natural long exact sequence*

$$\cdots \to \pi_k^s(A) \to \pi_k^s(X) \to \pi_k^s(X/A) \xrightarrow{\partial} \pi_{k-1}^s(A) \to \pi_{k-1}^s(X) \to \cdots.$$

(4) Additivity: *the inclusions induce an isomorphism $\pi_k^s(X) \oplus \pi_k^s(Y) \cong \pi_k^s(X \vee Y)$.*

In other words, $X \mapsto \pi_*^s(X)$ is a reduced homology theory.

Proof. Part (1) is clear. For (2), both groups are colimits of the same sequence of groups $\pi_{k+m+1}(\Sigma^{m+1}X)$, merely indexed differently.

For (3), let $X \cup CA$ be the reduced mapping cone of the inclusion. Since CA is a contractible subcomplex, collapsing it gives a homotopy equivalence $X \cup CA \simeq X/A$ [Hat02, Prop. 0.17]. Iterating the mapping cone construction gives the Puppe sequence

$$A \to X \to X \cup CA \to \Sigma A \to \Sigma X \to \Sigma(X \cup CA) \to \Sigma^2 A \to \cdots.$$

In this sequence each map is, up to homotopy equivalence and sign, the quotient of a CW complex by a subcomplex that is the image of the previous map [May99, Ch. 8]. For example, the map $X \cup CA \to \Sigma A$ is the collapse of X. Applying Proposition 7.3 at each spot shows that π_k^s of this sequence is exact. Rewriting $\pi_k^s(\Sigma^j A) \cong \pi_{k-j}^s(A)$ by part (2) gives the long exact sequence.

For (4), apply (3) to the pair $(X \vee Y, X)$, with quotient Y. The inclusion $X \to X \vee Y$ has a retraction, and so does the quotient map $X \vee Y \to Y$. Hence every ∂ vanishes, and the resulting short exact sequences $0 \to \pi_k^s(X) \to \pi_k^s(X \vee Y) \to \pi_k^s(Y) \to 0$ are split by the inclusion of Y. □

Theorem 7.4 says that π_*^s satisfies all the Eilenberg–Steenrod axioms for a reduced homology theory except the *dimension axiom*. Its value on S^0 is $\pi_k^s(S^0) = \pi_k^s$, which is not concentrated in degree 0. Stable homotopy is the most fundamental example of a *generalized* homology theory, and its "coefficients" are the stable stems. Its relationship to ordinary homology is as close as one could hope in low degrees.

Proposition 7.5 (Stable Hurewicz theorem)**.** *Let X be an $(n-1)$-connected CW complex with $n \geq 0$. Then $\pi_k^s(X) = 0$ for $k < n$, and the Hurewicz map gives an isomorphism $\pi_n^s(X) \cong \tilde{H}_n(X;\mathbb{Z})$.*

Proof. Choose $m \geq 2$ large enough for (7.1) to hold for $k = n$. The space $\Sigma^m X$ is $(n+m-1)$-connected, so $\pi_{k+m}(\Sigma^m X) = 0$ for $k < n$. Since $n + m \geq 2$, the Hurewicz theorem [Hat02, Thm. 4.32] gives $\pi_{n+m}(\Sigma^m X) \cong \tilde{H}_{n+m}(\Sigma^m X)$. The suspension isomorphism of homology identifies the latter with $\tilde{H}_n(X)$. □

For instance, $\pi_0^s(X) \cong \tilde{H}_0(X;\mathbb{Z})$ for every X, and $\pi_k^s(S^n) \cong \pi_{k-n}^s$. Rationally the two theories agree completely. Serre showed that the rational Hurewicz map gives $\pi_k^s(X) \otimes \mathbb{Q} \cong \tilde{H}_k(X;\mathbb{Q})$ [Ser53]. Thus everything that distinguishes stable homotopy from homology is torsion, and all of it is encoded in the groups π_k^s.

7.3. **The stable stems.** The most important case is $X = S^0$, for which $\Sigma^m S^0 = S^m$.

Definition 7.6. The *k-th stable stem* is $\pi_k^s = \pi_k^s(S^0) = \operatorname{colim}_m \pi_{k+m}(S^m)$.

Since S^m is $(m-1)$-connected, the map $\pi_{k+m}(S^m) \to \pi_{k+m+1}(S^{m+1})$ is an isomorphism when $k + m < 2m - 1$, i.e., $m > k + 1$, and a surjection when $m = k + 1$. Thus

$$\pi_k^s \cong \pi_{n+k}(S^n) \qquad \text{for all } n \geq k+2.$$

Our results so far give the first two stems:

- $\pi_0^s \cong \mathbb{Z}$, generated by the identity (Corollary 6.7, or Proposition 7.5);
- $\pi_1^s \cong \pi_4(S^3)$ is cyclic, generated by the image η of the Hopf map (Corollary 6.10). In fact it is $\mathbb{Z}/2$.

Table 1 shows the groups $\pi_{n+k}(S^n)$ for small n and k, together with the region where they have stabilized. Table 2 lists the first sixteen stems.

	S^1	S^2	S^3	S^4	S^5	S^6	S^7	S^8
π_n	$\mathbb{Z}$	$\mathbb{Z}$	$\mathbb{Z}$	$\mathbb{Z}$	$\mathbb{Z}$	$\mathbb{Z}$	$\mathbb{Z}$	$\mathbb{Z}$
π_{n+1}	0	$\mathbb{Z}$	$\mathbb{Z}/2$	$\mathbb{Z}/2$	$\mathbb{Z}/2$	$\mathbb{Z}/2$	$\mathbb{Z}/2$	$\mathbb{Z}/2$
π_{n+2}	0	$\mathbb{Z}/2$	$\mathbb{Z}/2$	$\mathbb{Z}/2$	$\mathbb{Z}/2$	$\mathbb{Z}/2$	$\mathbb{Z}/2$	$\mathbb{Z}/2$
π_{n+3}	0	$\mathbb{Z}/2$	$\mathbb{Z}/12$	$\mathbb{Z} \oplus \mathbb{Z}/12$	$\mathbb{Z}/24$	$\mathbb{Z}/24$	$\mathbb{Z}/24$	$\mathbb{Z}/24$
π_{n+4}	0	$\mathbb{Z}/12$	$\mathbb{Z}/2$	$(\mathbb{Z}/2)^2$	$\mathbb{Z}/2$	0	0	0
π_{n+5}	0	$\mathbb{Z}/2$	$\mathbb{Z}/2$	$(\mathbb{Z}/2)^2$	$\mathbb{Z}/2$	$\mathbb{Z}$	0	0
π_{n+6}	0	$\mathbb{Z}/2$	$\mathbb{Z}/3$	$\mathbb{Z}/24 \oplus \mathbb{Z}/3$	$\mathbb{Z}/2$	$\mathbb{Z}/2$	$\mathbb{Z}/2$	$\mathbb{Z}/2$
π_{n+7}	0	$\mathbb{Z}/3$	$\mathbb{Z}/15$	$\mathbb{Z}/15$	$\mathbb{Z}/30$	$\mathbb{Z}/60$	$\mathbb{Z}/120$	$\mathbb{Z} \oplus \mathbb{Z}/120$

TABLE 1. The groups $\pi_{n+k}(S^n)$ for $1 \leq n \leq 8$ and $0 \leq k \leq 7$ [Tod62], [Hat02, §4.1]. Shaded entries lie in the stable range $n \geq k + 2$ guaranteed by Theorem 6.5.

Remark 7.7. Several features of Table 1 are explained by results in this paper:

- The columns for S^2 and S^3 agree from π_3 on (Theorem 5.10).
- The S^4 column satisfies $\pi_{4+k}(S^4) \cong \pi_{4+k}(S^7) \oplus \pi_{3+k}(S^3)$ (Theorem 5.12). For example, $\pi_7(S^4) \cong \mathbb{Z} \oplus \mathbb{Z}/12$ and $\pi_{10}(S^4) \cong \mathbb{Z}/24 \oplus \mathbb{Z}/3$.
- The S^8 column, likewise, gives $\pi_{15}(S^8) \cong \pi_{15}(S^{15}) \oplus \pi_{14}(S^7) \cong \mathbb{Z} \oplus \mathbb{Z}/120$.
- Along each row, starting from $n = k+1$, the suspension maps are a surjection followed by isomorphisms, as Theorem 6.5 predicts. Before that point no such pattern holds: in the row $k = 3$, for example, $\Sigma_*\colon \pi_5(S^2) \cong \mathbb{Z}/2 \to \pi_6(S^3) \cong \mathbb{Z}/12$ is not surjective.

The infinite groups off the diagonal, $\pi_3(S^2)$, $\pi_7(S^4)$, $\pi_{11}(S^6)$, and $\pi_{15}(S^8)$, are no accident. Serre proved that $\pi_i(S^n)$ is finite except when $i = n$, or when n is even and $i = 2n-1$, in which case $\pi_{2n-1}(S^n) \cong \mathbb{Z} \oplus (\text{finite})$ [Ser53]. In particular every stable stem π_k^s with $k > 0$ is finite.

k	0	1	2	3	4	5	6	7
π_k^s	$\mathbb{Z}$	$\mathbb{Z}/2$	$\mathbb{Z}/2$	$\mathbb{Z}/24$	0	0	$\mathbb{Z}/2$	$\mathbb{Z}/240$
k	8	9	10	11	12	13	14	15
π_k^s	$(\mathbb{Z}/2)^2$	$(\mathbb{Z}/2)^3$	$\mathbb{Z}/6$	$\mathbb{Z}/504$	0	$\mathbb{Z}/3$	$(\mathbb{Z}/2)^2$	$\mathbb{Z}/480 \oplus \mathbb{Z}/2$

TABLE 2. The stable stems π_k^s for $0 \le k \le 15$ [Tod62, Rav86].

Primary decomposition. Since π_k^s is finite for $k > 0$, it splits as the direct sum of its p-primary parts. The p-primary parts have a striking pattern. For an odd prime p, the p-primary part of π_k^s vanishes for $0 < k < 2p-3$, and is $\mathbb{Z}/p$ for $k = 2p-3$ [Ser53, Tod62]. Table 2 confirms this: the first 3-torsion is in $\pi_3^s \cong \mathbb{Z}/24$, the first 5-torsion in $\pi_7^s \cong \mathbb{Z}/240$, and the first 7-torsion in $\pi_{11}^s \cong \mathbb{Z}/504$ (note $504 = 2^3 \cdot 3^2 \cdot 7$). The first 11-torsion appears in $\pi_{19}^s \cong \mathbb{Z}/264 \oplus \mathbb{Z}/2$, and $264 = 2^3 \cdot 3 \cdot 11$. In practice, the stems are computed one prime at a time.

7.4. **The ring of stable stems.** Stable stems can be composed. Let $\alpha \in \pi_p^s$ and $\beta \in \pi_q^s$. Represent α by $f\colon S^{p+m} \to S^m$ and, suspending if necessary, represent β by $g\colon S^{p+q+m} \to S^{p+m}$. Define

$$\alpha\beta = [f \circ g] \in \pi_{p+q}^s.$$

This is well defined, because suspending f and g suspends $f \circ g$.

Proposition 7.8. *With this product, $\pi_*^s = \bigoplus_{k \ge 0} \pi_k^s$ is a graded ring with unit $1 = [\mathrm{id}_{S^0}]$, and it is graded-commutative:*

$$\alpha\beta = (-1)^{pq}\beta\alpha \qquad (\alpha \in \pi_p^s,\ \beta \in \pi_q^s).$$

Sketch. Composition is always additive in the first variable: $(f_1 + f_2) \circ g = f_1 \circ g + f_2 \circ g$. It is additive in the second variable when the first map is a suspension, which can always be arranged stably. For commutativity, one checks that $f \circ g$ is stably homotopic to the smash product $f \wedge g\colon S^{p+m} \wedge S^{q+m'} \to S^m \wedge S^{m'}$. The two smash products $f \wedge g$ and $g \wedge f$ differ by composing on each side with the maps that interchange two smash factors of spheres. Interchanging S^a and S^b has degree $(-1)^{ab}$, so with m, m' even the total sign is

$$(-1)^{(p+m)(q+m')}(-1)^{mm'} = (-1)^{pq}.$$

See [Tod62, Ch. I] for details. □

The ring structure organizes Table 2 considerably. In low degrees [Tod62]:

- $2\eta = 0$;
- η^2 generates π_2^s;
- $\eta^3 = 12\nu$ is the element of order 2 in $\pi_3^s \cong \mathbb{Z}/24$, where ν is a generator;
- $\eta\nu = 0$, since $\pi_4^s = 0$;
- ν^2 generates π_6^s;
- if σ generates π_7^s, then $\sigma^2 \neq 0$ in π_{14}^s.

Since $\pi_4^s = 0$, we also get $\eta^4 = 0$. This is an instance of a general theorem of Nishida [Nis73]: every element of π_k^s with $k > 0$ is nilpotent. So the ring π_*^s is $\mathbb{Z}$ in degree 0 together with a nilpotent ideal of torsion elements.

8. A Glimpse of Stable Homotopy Theory

Section 7 showed that, after stabilization, homotopy groups behave like a homology theory. Stable homotopy theory is the study of the world in which this becomes the basic principle. In this final section we sketch its main structures, and show how they explain the patterns in Table 2. Proofs are mostly beyond our scope; we give references throughout, and [Ada74, Rav86] are good places to start.

8.1. **The Spanier–Whitehead category.** The homotopy groups $\pi_k^s(X)$ are the stable version of maps from spheres. More generally, for finite pointed CW complexes X and Y, define the group of *stable maps*

$$\{X, Y\} = \operatorname{colim}_m [\Sigma^m X, \Sigma^m Y].$$

A version of the Freudenthal theorem for maps shows that $[\Sigma^m X, \Sigma^m Y] \to [\Sigma^{m+1} X, \Sigma^{m+1} Y]$ is a bijection once $m > \dim X + 1$ [SW55], so this colimit is attained. These groups are the morphisms of the *Spanier–Whitehead category*. Its objects are pairs (X, n), with X a finite CW complex and $n \in \mathbb{Z}$, which we think of as formal desuspensions $\Sigma^n X$. Morphisms are $\{(X, n), (Y, n')\} = \operatorname{colim}_m [\Sigma^{m+n} X, \Sigma^{m+n'} Y]$. This category has several good properties that the homotopy category of spaces lacks:

- It is additive: each $\{X, Y\}$ is an abelian group, and composition is bilinear.
- Suspension is invertible, with inverse $(X, n) \mapsto (X, n-1)$. So there are spheres S^n for all $n \in \mathbb{Z}$, and $\{S^k, S^0\} = \pi_k^s$ for every integer k.
- A cofibre sequence $A \to X \to X/A$ induces long exact sequences of both $\{W, -\}$ and $\{-, W\}$. The first is Theorem 7.4 again; the second already holds unstably.

Exactness in both variables is the hallmark of the stable world. The category also has a duality. If X is a finite subcomplex of S^N, the complement $S^N \setminus X$ deformation retracts onto a finite complex, and

$$DX = \Sigma^{1-N}\big(S^N \setminus X\big)$$

is independent of N and of the embedding. This *Spanier–Whitehead dual* satisfies $D(DX) \simeq X$, $DS^n = S^{-n}$, and $\{X \wedge W, Y\} \cong \{W, DX \wedge Y\}$ [SW55]. On homology it realizes Alexander duality: $\tilde{H}_k(DX) \cong \tilde{H}^{-k}(X)$. So stable homotopy theory treats homology and cohomology symmetrically.

8.2. **Spectra.** The Spanier–Whitehead category has one serious defect: it contains only finite complexes. The spaces that represent cohomology, such as the Eilenberg–MacLane spaces $K(A, n)$ with $\tilde{H}^n(X; A) \cong [X, K(A, n)]$ [Hat02, Thm. 4.57], are infinite complexes. The objects that fix this are spectra.

Definition 8.1. A *spectrum* E is a sequence of pointed CW complexes $E_0, E_1, E_2, \dots$ with structure maps $\Sigma E_n \to E_{n+1}$. Its homotopy groups are

$$\pi_k(E) = \operatorname{colim}_n \pi_{k+n}(E_n), \qquad k \in \mathbb{Z}.$$

E is an Ω-*spectrum* if the adjoint maps $E_n \to \Omega E_{n+1}$ are weak homotopy equivalences.

Example 8.2.

(1) *Suspension spectra.* For a pointed CW complex X, the suspension spectrum $\Sigma^\infty X$ has $(\Sigma^\infty X)_n = \Sigma^n X$, with the identity maps as structure maps. By definition, $\pi_k(\Sigma^\infty X) = \pi_k^s(X)$. The *sphere spectrum* is $\mathbb{S} = \Sigma^\infty S^0$, with $\pi_k(\mathbb{S}) = \pi_k^s$.
(2) *Eilenberg–MacLane spectra.* For an abelian group A, the spectrum HA has $(HA)_n = K(A, n)$. The structure maps are adjoint to the weak equivalences $K(A, n) \simeq \Omega K(A, n+1)$. Then $\pi_0(HA) = A$, and all other homotopy groups vanish.
(3) *Complex K-theory.* Bott periodicity [Bot59] gives homotopy equivalences $\Omega U \simeq \mathbb{Z} \times BU$ and $\Omega(\mathbb{Z} \times BU) \simeq U$. The spectrum KU with $KU_{2n} = \mathbb{Z} \times BU$ and $KU_{2n+1} = U$ is therefore an Ω-spectrum, with $\pi_k(KU) = \mathbb{Z}$ for k even and 0 for k odd.
(4) *Thom spectra.* The Thom spaces of the universal vector bundles over $BO(n)$ form a spectrum MO. Similarly, the universal complex bundles over $BU(n)$ give a spectrum MU. Thom [Tho54] computed $\pi_*(MO)$: it is the unoriented cobordism ring, a polynomial ring over $\mathbb{F}_2$ on generators x_i, one in each degree $i \neq 2^j - 1$. Milnor and Novikov showed that $\pi_*(MU) \cong \mathbb{Z}[x_1, x_2, \dots]$ with $|x_i| = 2i$ [Rav86].

The comparison between MO and $\mathbb{S}$ is instructive. Thom spectra have enormous, well-understood homology, and that homology determines their homotopy completely. The sphere spectrum has the smallest possible homology, $\mathbb{Z}$ in degree 0, yet its homotopy is the most complicated object in the subject.

Every spectrum E defines a homology theory and a cohomology theory on pointed CW complexes,

$$E_k(X) = \operatorname{colim}_n \pi_{k+n}(E_n \wedge X), \qquad E^k(X) = \operatorname{colim}_n [\Sigma^n X, E_{k+n}],$$

by a theorem of G. W. Whitehead [Whi62]. Examples:

- the sphere spectrum gives $\mathbb{S}_k(X) = \pi_k^s(X)$;
- $H\mathbb{Z}$ gives ordinary homology and cohomology;
- KU gives complex K-theory;
- MO and MU give unoriented and complex bordism.

Conversely, the Brown representability theorem [Bro62], [Hat02, §4.E] says that every reduced cohomology theory on pointed CW complexes is of the form E^* for some Ω-spectrum E. Spectra are thus to cohomology theories what spaces are to homotopy types.

Spectra form a *stable homotopy category*. It contains the Spanier–Whitehead category as the full subcategory of finite spectra, Σ is invertible in it, and it has a smash product with unit $\mathbb{S}$. Constructing that smash product carefully is subtle. It was first done by Boardman, whose stable homotopy category is the one presented in [Ada74]. Modern foundations include symmetric spectra [HSS00], S-modules [EKMM97], and stable ∞-categories [Lur17]. In this language $\pi_k^s = [\Sigma^k \mathbb{S}, \mathbb{S}]$, and π_*^s is the ring of graded endomorphisms of the unit object. Every $E_*(X)$ is a module over π_*^s. This is the precise sense in which the stable stems are the coefficients of all of stable homotopy theory.

8.3. **Framed manifolds.** The stable stems have a purely geometric description, due to Pontryagin and Thom. A *framed* k-manifold in $\mathbb{R}^{n+k}$ is a closed submanifold $M^k \subseteq \mathbb{R}^{n+k}$ together with a trivialization $\nu_M \cong M \times \mathbb{R}^n$ of its normal bundle. Two framed k-manifolds are *framed cobordant* if together they form the boundary of a framed $(k+1)$-manifold in $\mathbb{R}^{n+k} \times I$. We write $\Omega_k^{\mathrm{fr}}(n)$ for the set of framed cobordism classes.

Theorem 8.3 (Pontryagin–Thom [Pon55], [Mil65, §7])**.** *For all n and k there is a bijection $\Omega_k^{\mathrm{fr}}(n) \cong \pi_{n+k}(S^n)$. It is compatible with the inclusions $\mathbb{R}^{n+k} \subseteq \mathbb{R}^{n+k+1}$ and with suspension, and so induces an isomorphism*

$$\Omega_k^{\mathrm{fr}} = \operatorname{colim}_n \Omega_k^{\mathrm{fr}}(n) \cong \pi_k^s.$$

Idea. Let $M \subseteq \mathbb{R}^{n+k}$ be framed. The framing identifies a tubular neighbourhood U of M with $M \times D^n$. The *collapse map*

$$S^{n+k} = \mathbb{R}^{n+k} \cup \{\infty\} \longrightarrow S^{n+k}/\big(S^{n+k} \setminus \operatorname{int} U\big) \cong M_+ \wedge \big(D^n/\partial D^n\big) \longrightarrow D^n/\partial D^n = S^n$$

sends everything outside U to the basepoint, and is the projection to the normal disc inside U. Conversely, a smooth map $f\colon S^{n+k} \to S^n$ has a regular value y by Sard's theorem. Then $f^{-1}(y)$ is a closed k-manifold, framed by pulling back a basis of $T_y S^n$. One checks that homotopies correspond to framed cobordisms. □

The low stems can be seen directly in this picture:

- $k = 0$. A framed 0-manifold is a finite set of points, each with an orientation sign. Framed cobordism classes are counted by the signed number of points, so $\Omega_0^{\mathrm{fr}} \cong \mathbb{Z}$. This is the degree, and recovers $\pi_n(S^n) \cong \mathbb{Z}$ (Corollary 6.7).
- $k = 1$. A framed 1-manifold is a union of circles. For $n \geq 3$, the framings of a circle $S^1 \subseteq \mathbb{R}^{n+1}$ form a torsor under $\pi_1(SO(n)) \cong \mathbb{Z}/2$: one framing extends over a disc, and the other, which comes from the Lie group structure of $S^1 = U(1)$, does not. The twisted circle represents η, and the fact that two twists cancel is the geometric content of $2\eta = 0$.
- $k = 3$. The sphere $S^3 = SU(2)$ with its left-invariant (Lie group) framing represents a generator ν of $\pi_3^s \cong \mathbb{Z}/24$.

A framed manifold of dimension $4m+2$ has a *Kervaire invariant* in $\mathbb{Z}/2$, the Arf invariant of a quadratic refinement of the intersection form on $H_{2m+1}(M; \mathbb{F}_2)$. Browder [Bro69] showed that it can be nonzero only in dimensions $2^{j+1} - 2$. Examples are known in dimensions $2, 6, 14, 30, 62$. Hill, Hopkins, and Ravenel [HHR16] showed that there are none in dimensions ≥ 254, and Lin, Wang, and Xu [LWX24] constructed one in dimension 126. This completes the answer: framed manifolds of Kervaire invariant one exist exactly in dimensions $2, 6, 14, 30, 62$, and 126. The examples in dimensions 2, 6, and 14 represent η^2, ν^2, and σ^2.

8.4. **The Hopf invariant and the J-homomorphism.** The Hopf maps of Section 5 come from multiplications on spheres, and this idea leads to the two classical families of elements of π_*^s.

The Hopf construction. Given a map $\mu\colon X \times Y \to Z$, define $H(\mu)\colon X * Y \to SZ$ on the join $X * Y$ by $[x, y, t] \mapsto [\mu(x, y), t]$. Since $S^p * S^q \cong S^{p+q+1}$, a map $\mu\colon S^{n-1} \times S^{n-1} \to S^{n-1}$ yields $H(\mu)\colon S^{2n-1} \to S^n$. Applied to multiplication in $\mathbb{C}$, $\mathbb{H}$, and the octonions $\mathbb{O}$, restricted to unit spheres, it yields the Hopf maps $\eta\colon S^3 \to S^2$, $\nu\colon S^7 \to S^4$, and $\sigma\colon S^{15} \to S^8$ [Hat02, §4.B].

The Hopf invariant. For $f\colon S^{2n-1} \to S^n$ with $n \geq 2$, the mapping cone $C_f = S^n \cup_f e^{2n}$ has integral cohomology generated by classes $a \in H^n$ and $b \in H^{2n}$. The *Hopf invariant*

$H(f) \in \mathbb{Z}$ is defined by $a^2 = H(f)\, b$. For n odd, graded commutativity gives $a^2 = -a^2$, so $H(f) = 0$. For the Hopf maps, the mapping cones are the projective planes $\mathbb{C}P^2$, $\mathbb{H}P^2$, and $\mathbb{O}P^2$, so $H = \pm 1$. For n even, the Whitehead product $[\iota_n, \iota_n]$ has Hopf invariant ± 2.

Theorem 8.4 (Adams [Ada60])**.** *A map $S^{2n-1} \to S^n$ of Hopf invariant one exists only for $n = 2, 4, 8$.*

Adams' theorem has well-known consequences. $\mathbb{R}^n$ admits a bilinear multiplication without zero divisors only for $n = 1, 2, 4, 8$. The sphere S^{n-1} admits a continuous multiplication with a two-sided unit only for $n = 1, 2, 4, 8$, and it is parallelizable only for $n = 1, 2, 4, 8$. Adams' proof is a tour de force with secondary cohomology operations. Adams and Atiyah later gave a short proof using Adams operations in K-theory [AA66].

The J-homomorphism. Let $\alpha\colon S^k \to SO(n)$. Then $\mu_\alpha(x, y) = \alpha(x)\, y$ defines $S^k \times S^{n-1} \to S^{n-1}$, and its Hopf construction is a map $J(\alpha)\colon S^{k+n} \to S^n$. This gives homomorphisms $J\colon \pi_k(SO(n)) \to \pi_{k+n}(S^n)$ [Whi42], compatible with the inclusions $SO(n) \subseteq SO(n+1)$ and with suspension. Passing to the colimit $SO = \operatorname{colim}_n SO(n)$ gives the *stable J-homomorphism*

$$J\colon \pi_k(SO) \to \pi_k^s.$$

For example, the generator of $\pi_1(SO(2)) \cong \mathbb{Z}$ is the rotation $x \mapsto (y \mapsto xy)$ of $\mathbb{C}$. Its μ is complex multiplication, so J of it is the Hopf map η. The source of J is known completely, by Bott periodicity [Bot59]: for $k \geq 1$,

$$\pi_k(SO) \cong \mathbb{Z}/2,\ \mathbb{Z}/2,\ 0,\ \mathbb{Z},\ 0,\ 0,\ 0,\ \mathbb{Z} \qquad \text{for } k \equiv 0, 1, 2, 3, 4, 5, 6, 7 \pmod 8.$$

Theorem 8.5 (Adams, Quillen [Ada66, Qui71])**.** *For $k \geq 1$, the image of $J\colon \pi_k(SO) \to \pi_k^s$ is:*

- *$\mathbb{Z}/2$ (J is injective) if $k \equiv 0, 1 \pmod 8$;*
- *cyclic of order equal to the denominator of $B_{2j}/4j$ if $k = 4j - 1$, where B_{2j} is the $2j$-th Bernoulli number;*
- *zero otherwise.*

Moreover, the image of J is detected by K-theory.

Table 3 compares the image of J with the stable stems. The denominators are 24, 240, 504, 480 for $j = 1, 2, 3, 4$, coming from $B_2 = \frac{1}{6}$, $B_4 = -\frac{1}{30}$, $B_6 = \frac{1}{42}$, $B_8 = -\frac{1}{30}$. The image of J accounts for all of π_k^s for $k = 1, 3, 7, 11$, and for a large part of it in general. Number theory enters homotopy theory here through the Bernoulli numbers, which govern the image of J in each degree $4j - 1$.

k	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
$\pi_k(SO)$	$\mathbb{Z}/2$	0	$\mathbb{Z}$	0	0	0	$\mathbb{Z}$	$\mathbb{Z}/2$	$\mathbb{Z}/2$	0	$\mathbb{Z}$	0	0	0	$\mathbb{Z}$
$\operatorname{im} J$	$\mathbb{Z}/2$	0	$\mathbb{Z}/24$	0	0	0	$\mathbb{Z}/240$	$\mathbb{Z}/2$	$\mathbb{Z}/2$	0	$\mathbb{Z}/504$	0	0	0	$\mathbb{Z}/480$
$\lvert\pi_k^s\rvert$	2	2	24	1	1	2	240	4	8	6	504	1	3	4	960

TABLE 3. The stable J-homomorphism in degrees $1 \leq k \leq 15$ (Theorem 8.5), compared with the orders of the stable stems (Table 2).

8.5. **Computing the stems: the Adams spectral sequence.** The systematic computation of the stable stems is done with spectral sequences. The most important is the Adams spectral sequence [Ada58]. At the prime 2 it has the form

$$E_2^{s,t} = \operatorname{Ext}_{\mathcal{A}}^{s,t}(\mathbb{F}_2, \mathbb{F}_2) \Longrightarrow \pi_{t-s}^s \otimes \mathbb{Z}_2,$$

where $\mathcal{A}$ is the mod 2 Steenrod algebra of stable cohomology operations, and $\mathbb{Z}_2$ denotes the 2-adic integers. For $k > 0$, the group $\pi_k^s \otimes \mathbb{Z}_2$ is simply the 2-primary part of π_k^s. The E_2 page is purely algebraic, computable by hand in a range and by computer far beyond. The differentials $d_r \colon E_r^{s,t} \to E_r^{s+r,t+r-1}$ are where the topology lives.

It is customary to draw the spectral sequence with the stem $t - s$ horizontally and the Adams filtration s vertically. Some basic features of the chart:

- The classes $h_i \in \operatorname{Ext}^{1,2^i}$, dual to the Steenrod squares Sq^{2^i}, lie in stem $2^i - 1$.
- The class h_0 detects 2, and h_1, h_2, h_3 detect η, ν, σ.
- Vertical lines denote multiplication by h_0, that is, by 2; diagonal lines denote multiplication by h_1, that is, by η.

Figure 2 shows the chart through stem 7, where there are no differentials, so $E_2 = E_\infty$.

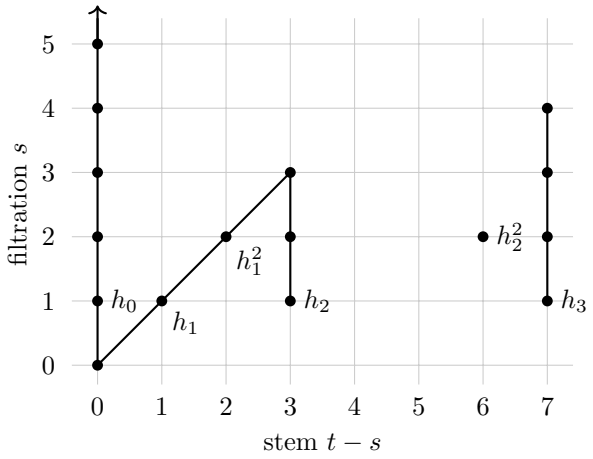


FIGURE 2. The Adams spectral sequence at $p = 2$ through stem 7. Vertical lines are multiplication by 2, diagonal lines by η. Reading off columns gives the 2-primary parts of the stems: $\mathbb{Z}_2$, $\mathbb{Z}/2$, $\mathbb{Z}/2$, $\mathbb{Z}/8$, 0, 0, $\mathbb{Z}/2$, $\mathbb{Z}/16$. Compare Table 2: $24 = 8 \cdot 3$ and $240 = 16 \cdot 15$. The relation $h_1^3 = h_0^2 h_2$ is $\eta^3 = 4\nu$ at the prime 2.

The Adams spectral sequence turns geometric questions into questions about specific classes:

- *Hopf invariant one.* An element of Hopf invariant one in $\pi_{2^{i+1}-1}(S^{2^i})$ exists if and only if h_i survives to E_∞. Adams proved Theorem 8.4 by showing that

$$d_2(h_i) = h_0 h_{i-1}^2 \neq 0 \qquad \text{for } i \geq 4.$$

- *Kervaire invariant one.* By Browder's theorem [Bro69], a framed manifold of Kervaire invariant one exists in dimension $2^{j+1} - 2$ if and only if the class h_j^2 survives. This is the form in which Hill–Hopkins–Ravenel and Lin–Wang–Xu attacked the problem.

Beyond the classical Adams spectral sequence, the Adams–Novikov spectral sequence replaces mod 2 cohomology by complex cobordism MU [Rav86]. Modern computations combine the two with motivic homotopy theory and machine computation. This is how Isaksen, Wang, and Xu determined the stems through dimension 90, apart from a few carefully listed uncertainties [IWX23].

8.6. **The chromatic picture.** Computing π_*^s one group at a time cannot be the end of the story, and the modern viewpoint looks instead for large-scale structure. Nishida's nilpotence theorem is a special case of the nilpotence theorem of Devinatz, Hopkins, and Smith [DHS88]. It says that a self-map of a finite spectrum is nilpotent if and only if it induces a nilpotent map on complex bordism MU_*. Together with the periodicity theorem of Hopkins and Smith [HS98], it organizes stable homotopy into layers indexed by a "height" $n \geq 0$. Each layer consists of periodic families of elements, governed by a cohomology theory of height n:

- Height 0 is rational homotopy: $\pi_*^s \otimes \mathbb{Q} = \mathbb{Q}$ in degree 0.
- Height 1 is governed by K-theory, and its periodic families recover the image of J (Theorem 8.5).
- Height 2 is governed by elliptic cohomology and topological modular forms.

This *chromatic* filtration [Rav92] is the organizing principle of much of contemporary research in stable homotopy theory.

The homotopy groups of spheres are still not completely known, and no one expects a closed formula. Asking how many essentially different maps $S^{n+k} \to S^n$ there are has led to spectral sequences, generalized cohomology theories, cobordism, K-theory, and connections to number theory. That such a simple question leads this deep is a good measure of how much structure homotopy theory contains.

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